

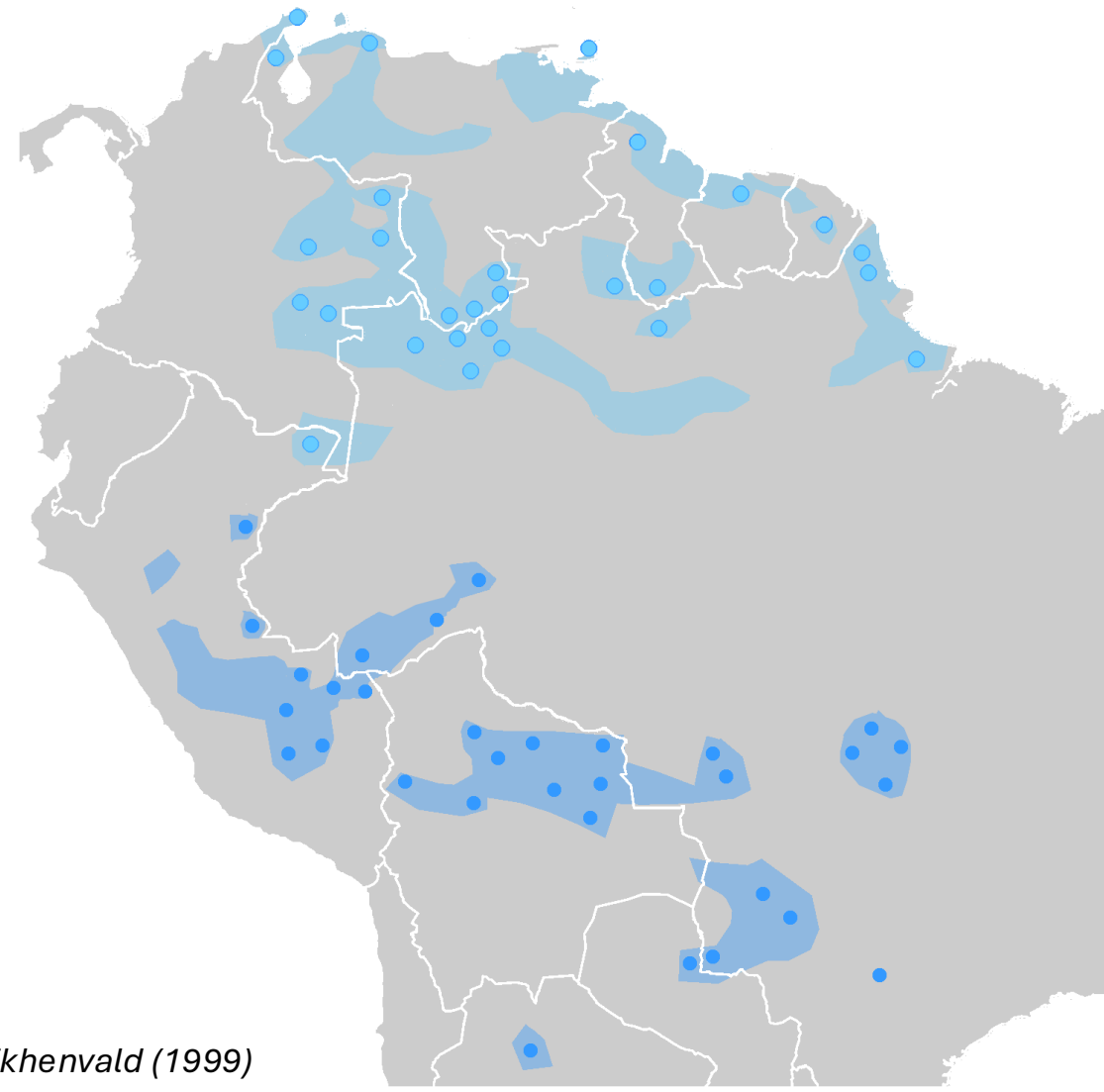
The Sounds of Migration

Computational Phonology
for Phylogenetic Inference
in Arawak Languages

Ian Jackson

Native American and Indigenous
Research Symposium

Similarities and Differences in the Arawak Language Family



via Wikipedia, with data from Aikhenvald (1999)

Similarities and Differences in the Arawak Language Family

Numeral	Garifuna	Wayuu	Kalinago	
1	Aban	Wane	Aba	
2	Biama	Piama	Biama	
3	Urüwa	Apünüin	Eroa	
4	Gádürün	Pienchi	Biámbüri	
5	Seingü	J'arai	Ábakabo	
6	Sisi	A'ipirua	Ábapuruko	
7	Sedü	Akaratchi	Biámpuruko	
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3	Urüwa	Apünüin	Eroa	Trois
4	Gádürün	Pienchi	Biámbüri	Quarte
5	Seingü	J'arai	Ábakabo	Cinq
6	Sisi	A'ipirua	Ábapuruko	Six
7	Sedü	Akaratchi	Biámpuruko	Sept
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Overlapping word forms

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Overlapping word forms

Variations in pronunciation

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Overlapping word forms

Variations in pronunciation

Influence from trading and colonization

Several influences on Arawak language diversity

Migration



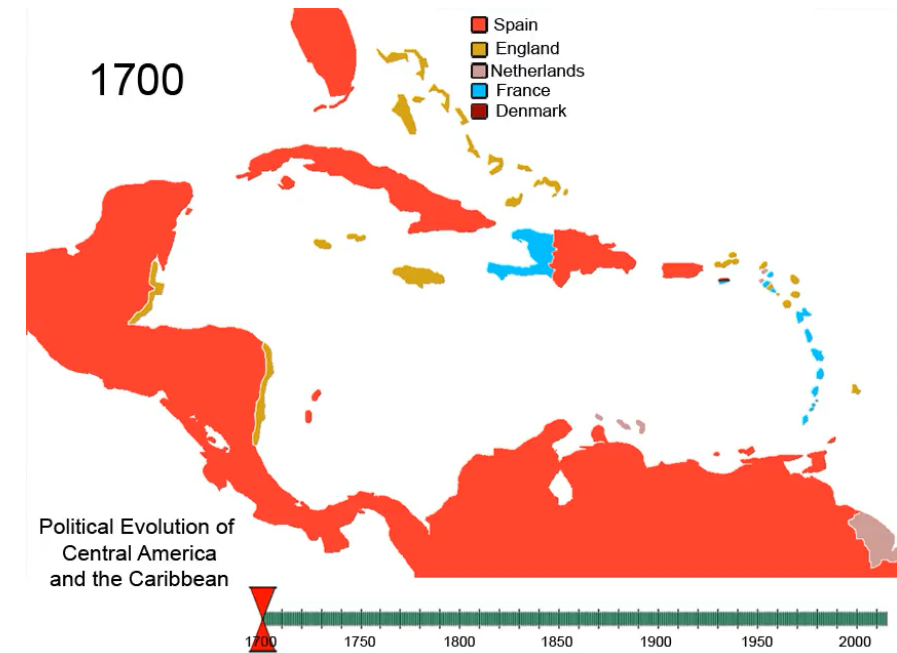
via YouTube

Trade



Chacon (2022)

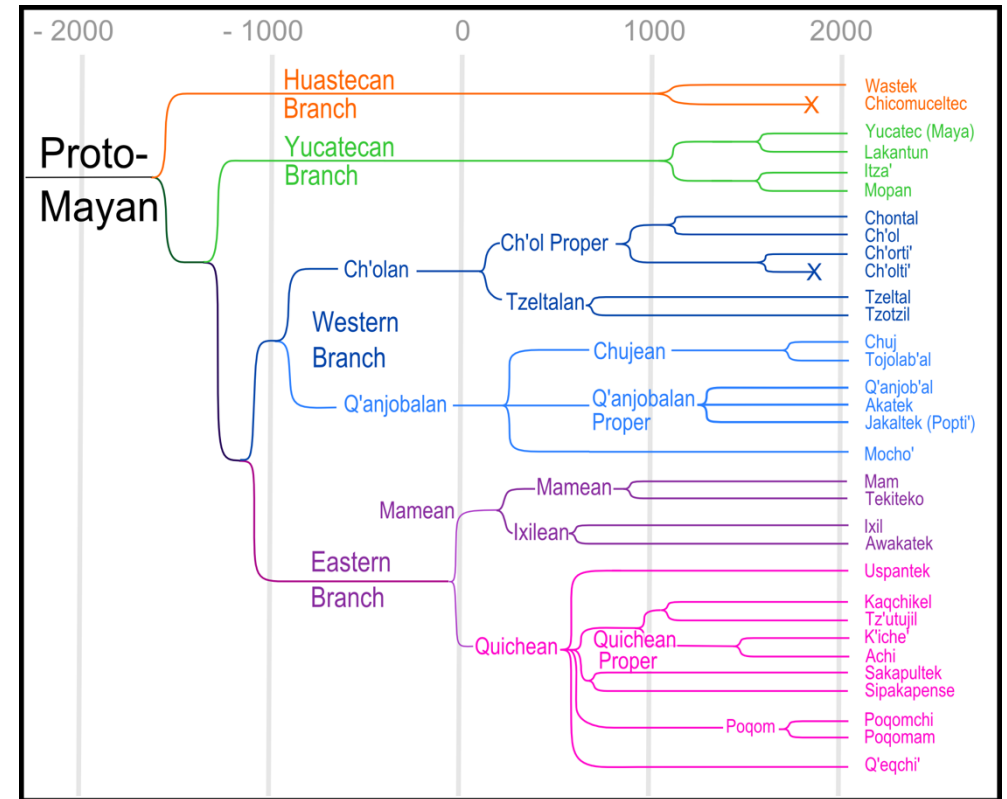
Colonization



via TheCleverTeacher

Computational Phylogenetics in Linguistics

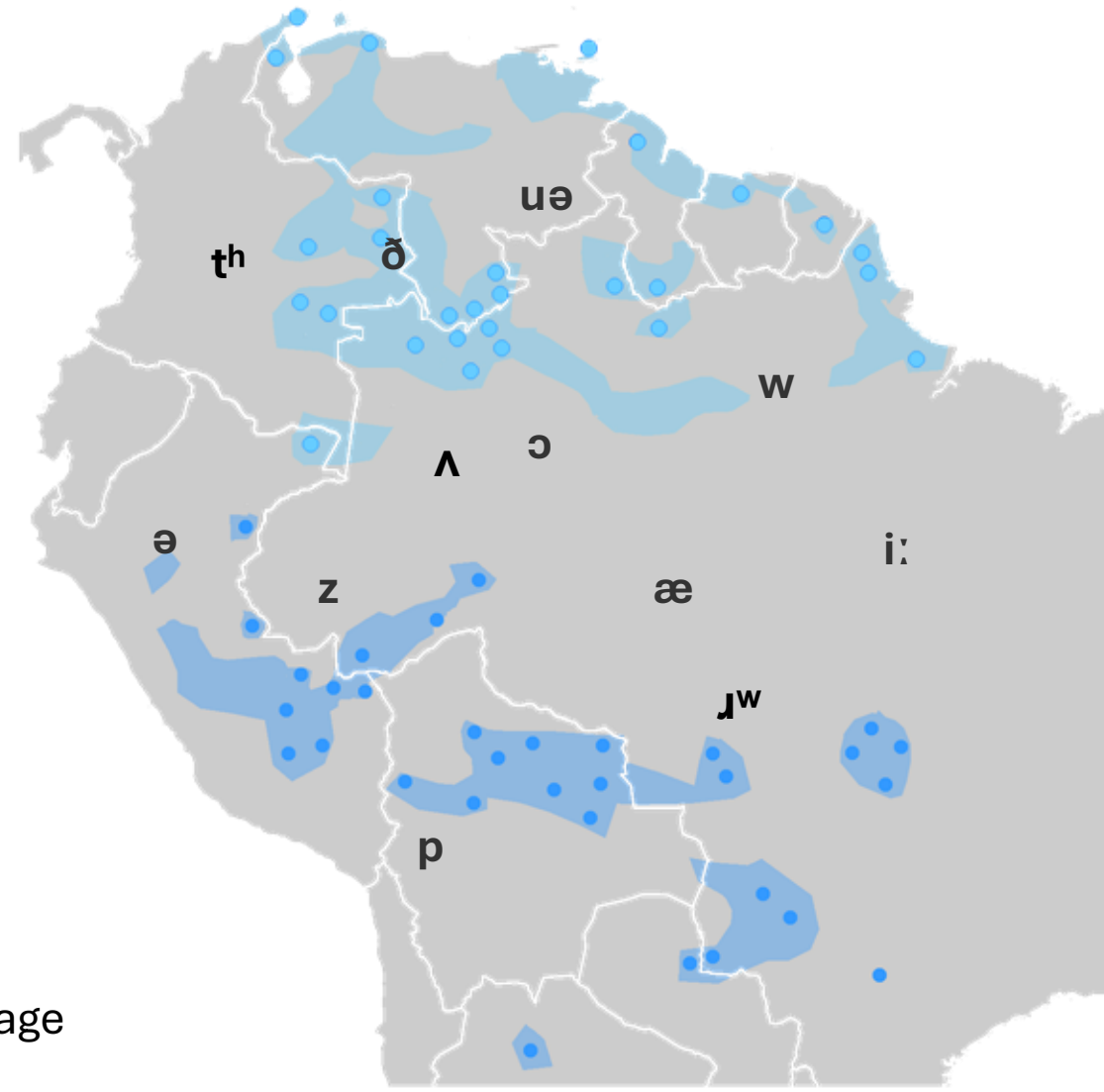
- Comparing languages to infer historical “family tree”
 - Typically done with word form or phonological (sound) comparison
- Difficult to do with indigenous (“low resource”) languages like Arawakan
 - Post-colonial erasure
 - Many oral (historically non-written) languages
 - Fewer orthographic norms
 - High linguistic diversity



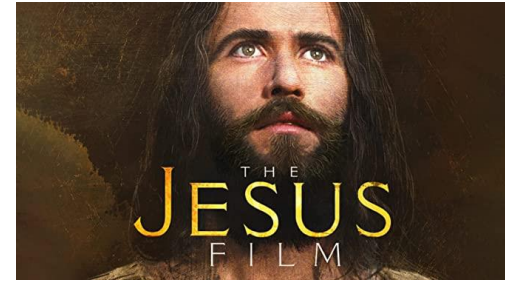
Computational Phylogenetics in Linguistics

- Previous studies of Arawak language family use phonological features to infer phylogenetic structure
 - Based on arbitrary boundaries, language-specific
 - Modern machine learning tools allow for **language-agnostic phone recognition**
- Project goal: Infer phylogenetic structure of the Arawak language family via neural network-guided phonological analysis

Phones = discrete sound units that constitute spoken language



Corpus and Languages Chosen



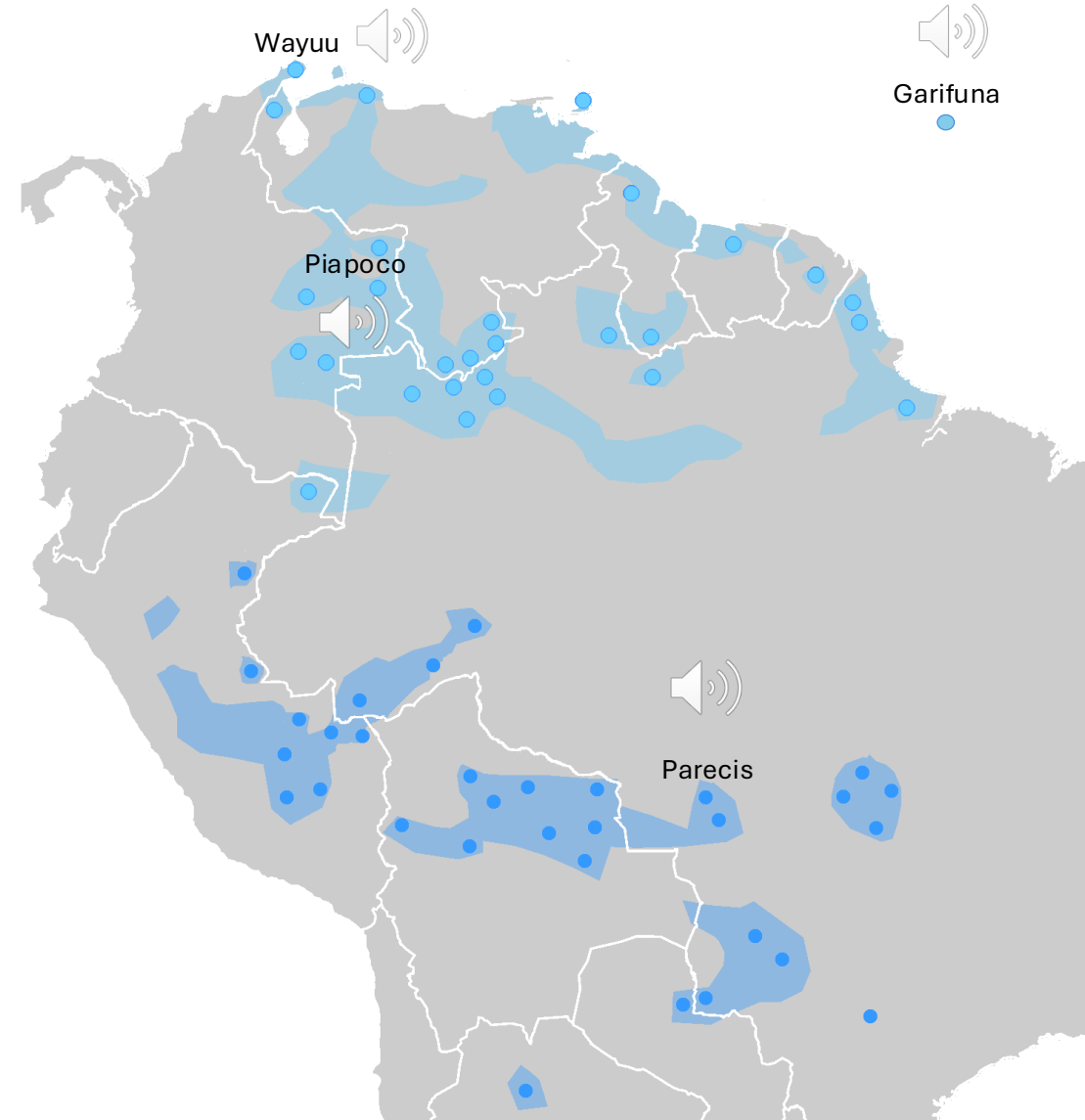
Jesus
Film®

- Corpus: Jesus (1979) via *The Jesus Film Project*

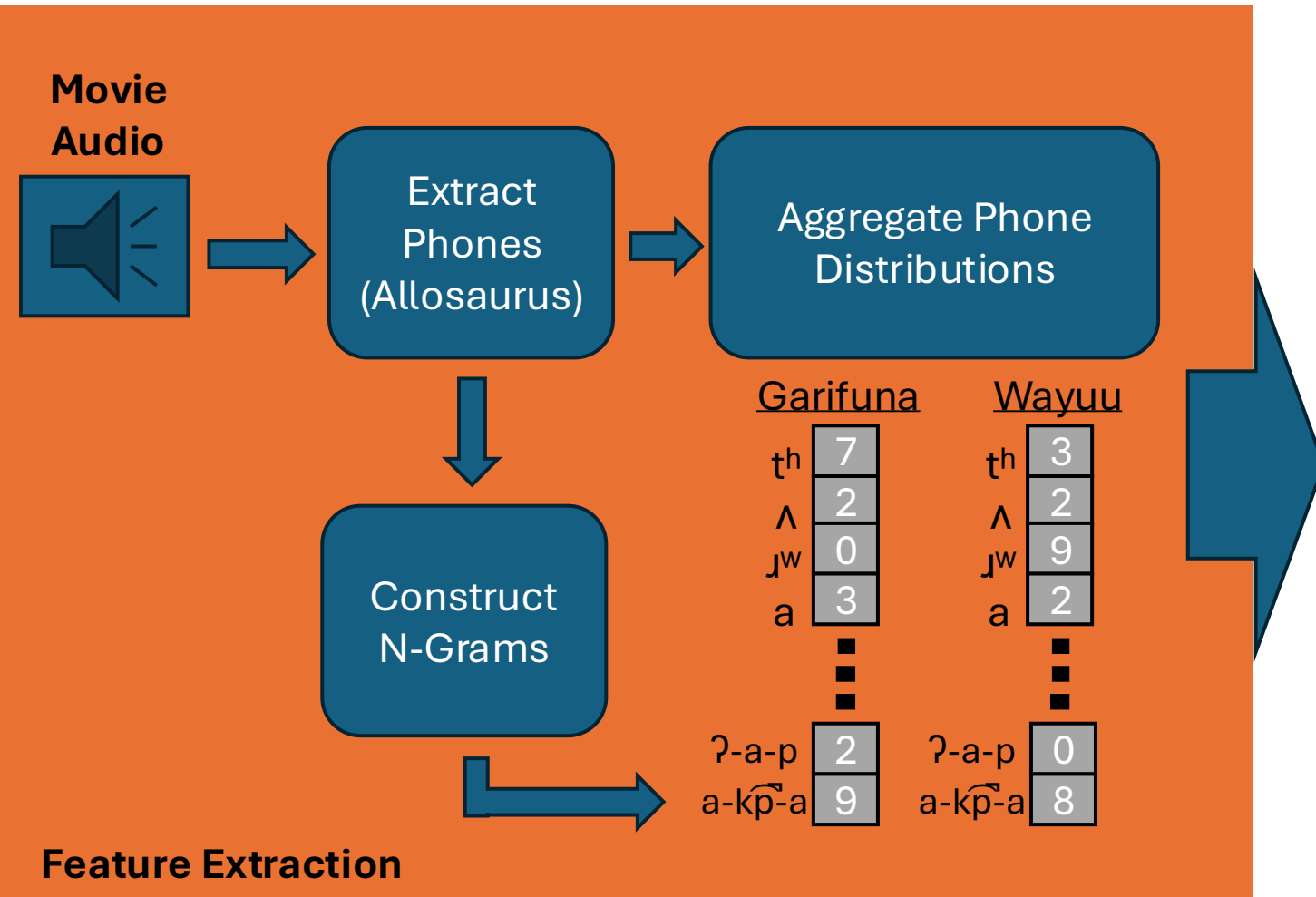


Corpus and Languages Chosen

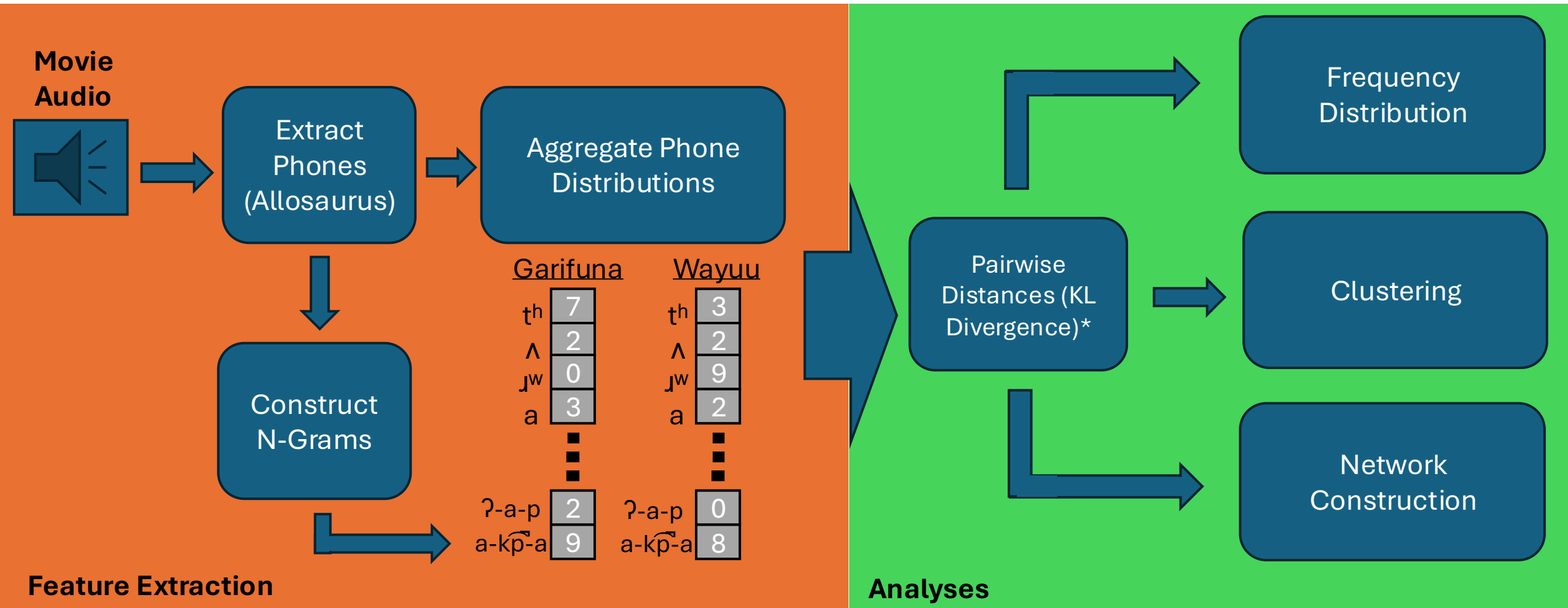
- Corpus: Jesus (1979) via *The Jesus Film Project*
- Arawak languages
 - Garifuna
 - Wayuu
 - Piapoco
 - Parecis
- Non-Arawak languages
 - Spanish
 - Portuguese
 - French
 - English



Phonological Analysis Pipeline

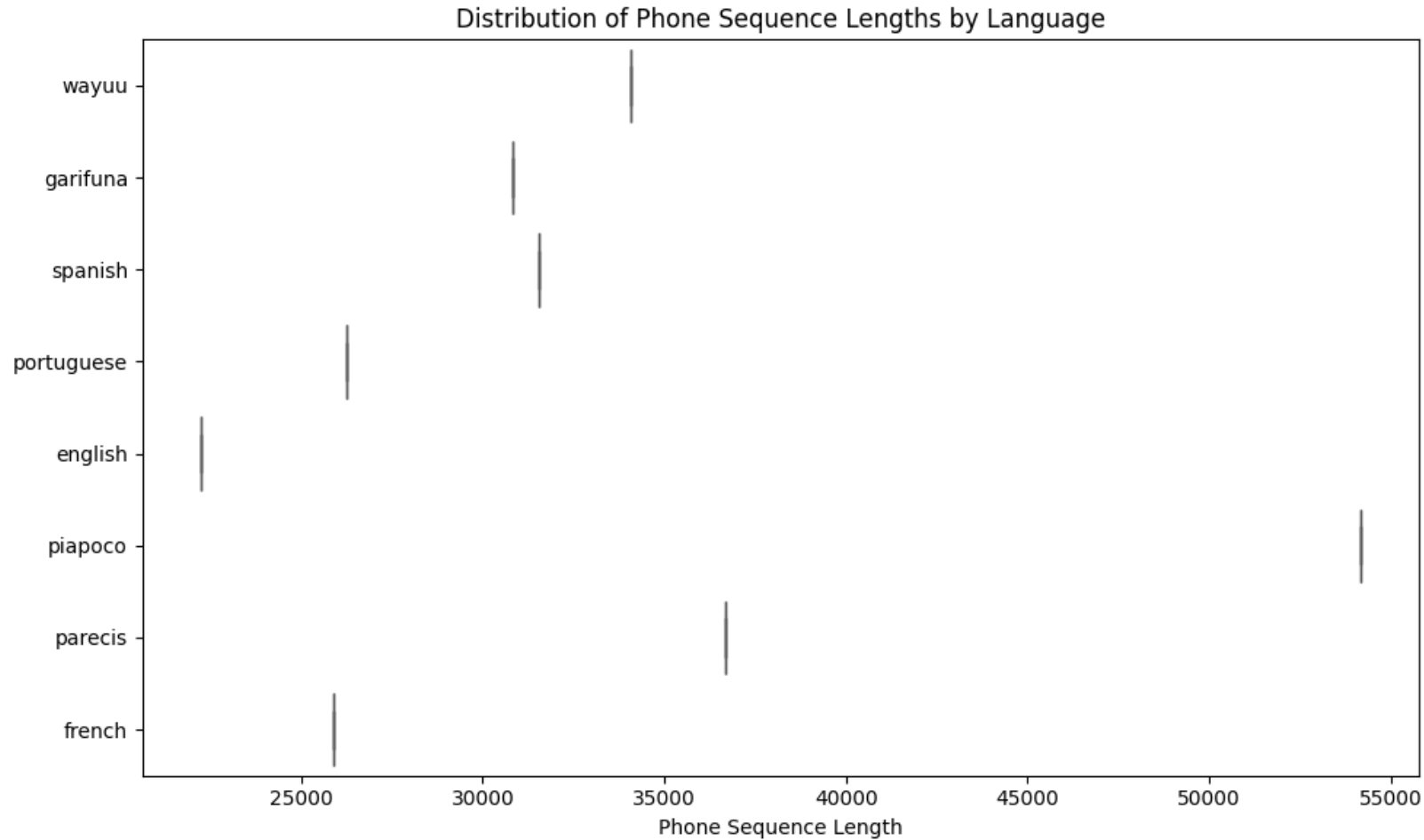


Phonological Analysis Pipeline



Results

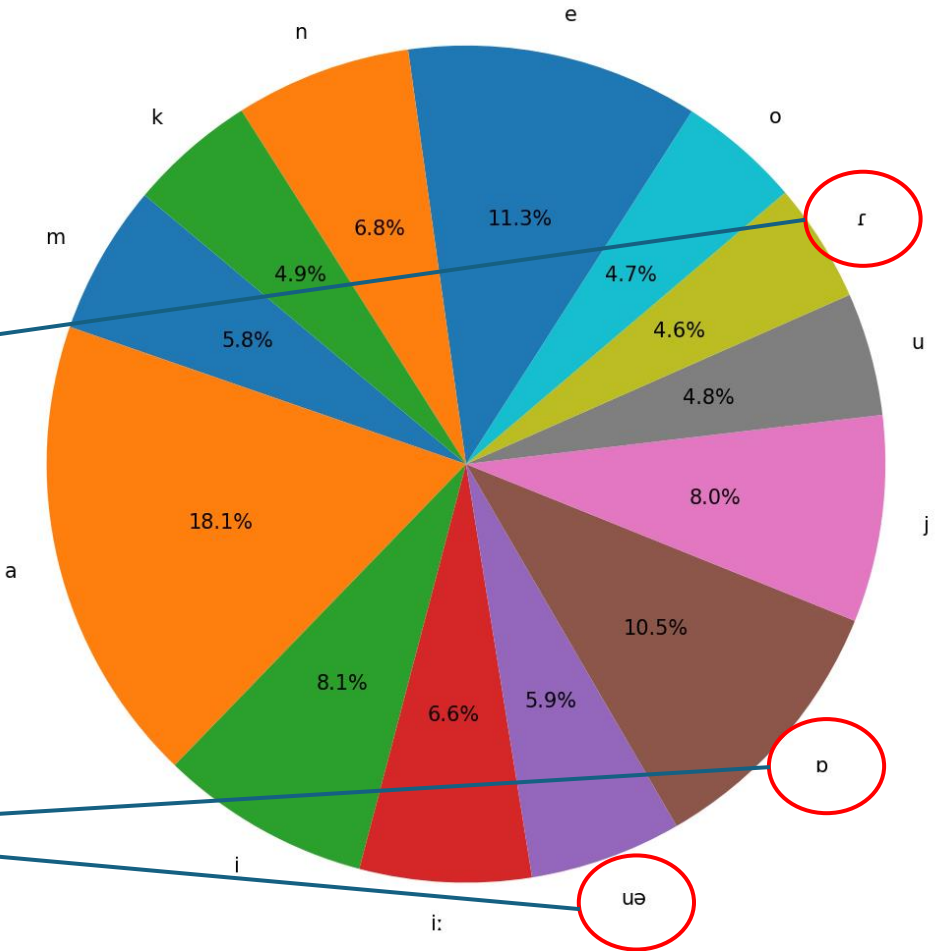
Languages vary by phonological frequency



Aggregated Arawak phonological distribution differs from Aikhenvald (1999)

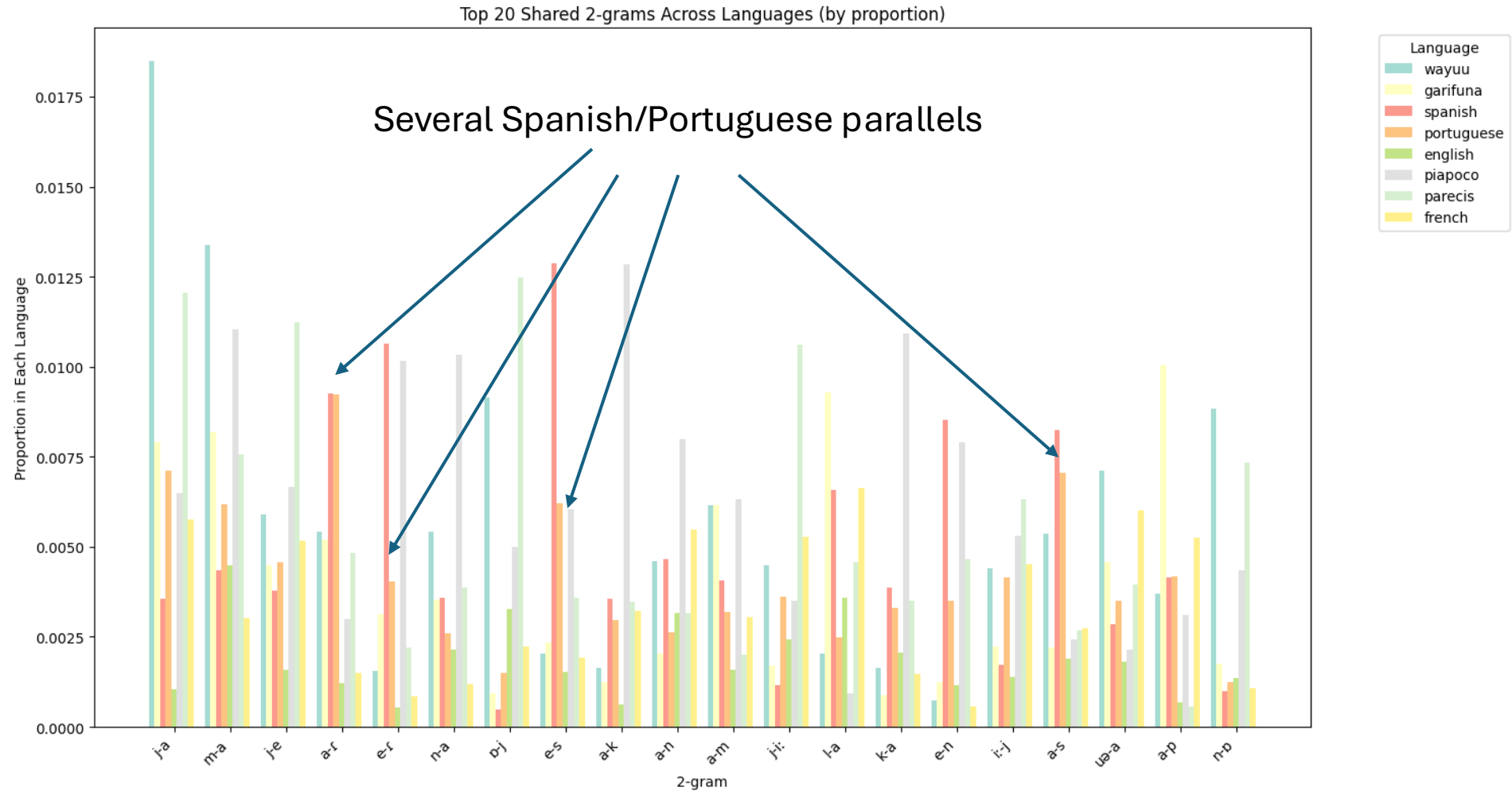
		Labial	Dental	Alveolar	Lamino-(alveo)-palatal	Velar	Glottal
Stop	voiced	(b)	d			g	
	voiceless	p	t			k	(ʔ)
	voiceless aspirated	(pʰ)	(tʰ)			(kʰ)	
Affricate				ts	tʃ		
Fricative		(ɸ)		s	ʃ	h	
Lateral				?	l		
Vibrant					r		
Nasal		m		n	ɲ		
Glide		w			j		

	Front	Central	Back
High	i i:	ɨ ɨ:	u u:
Mid		e e:	
Low		a a:	

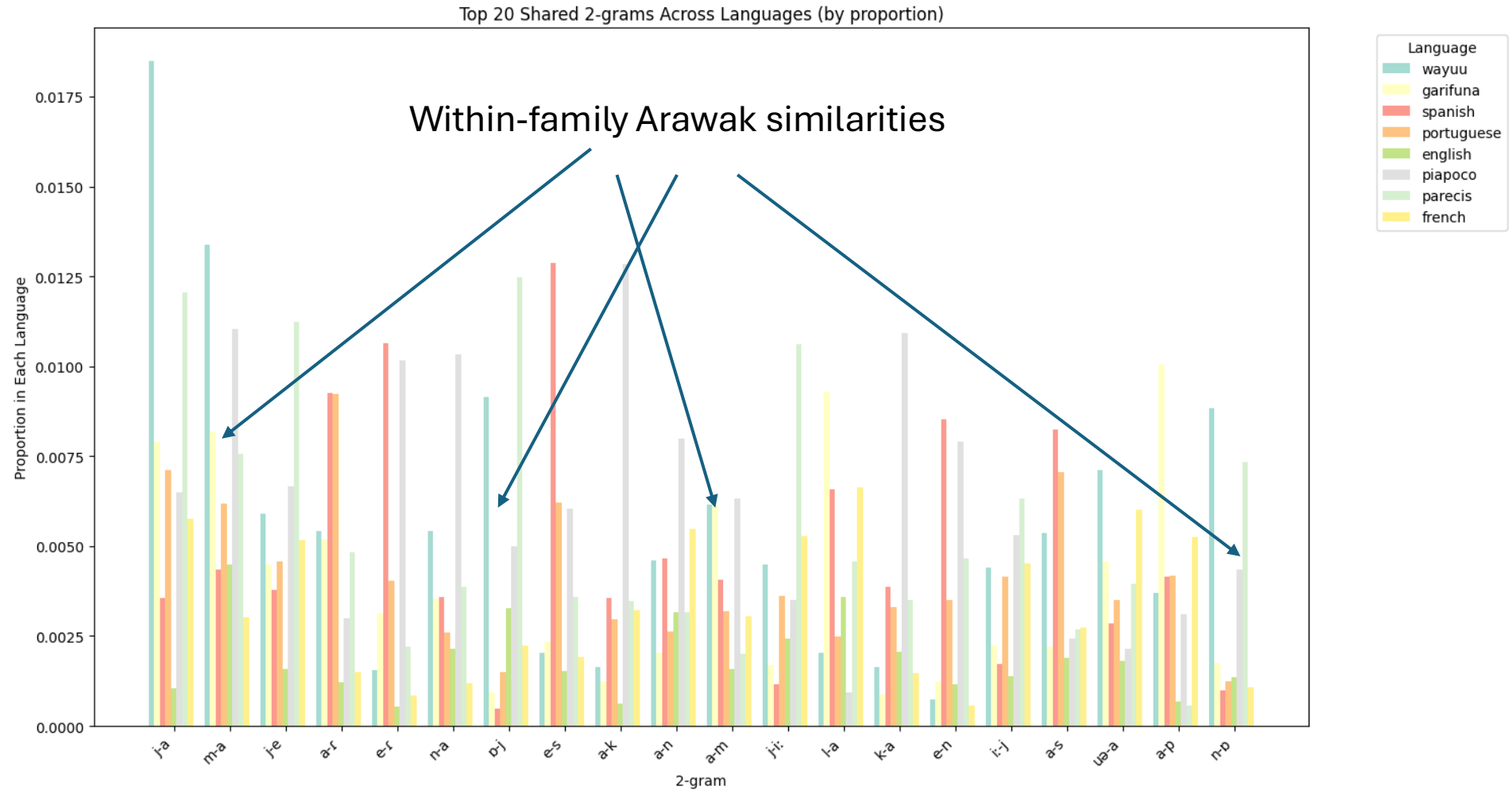


via Wikipedia

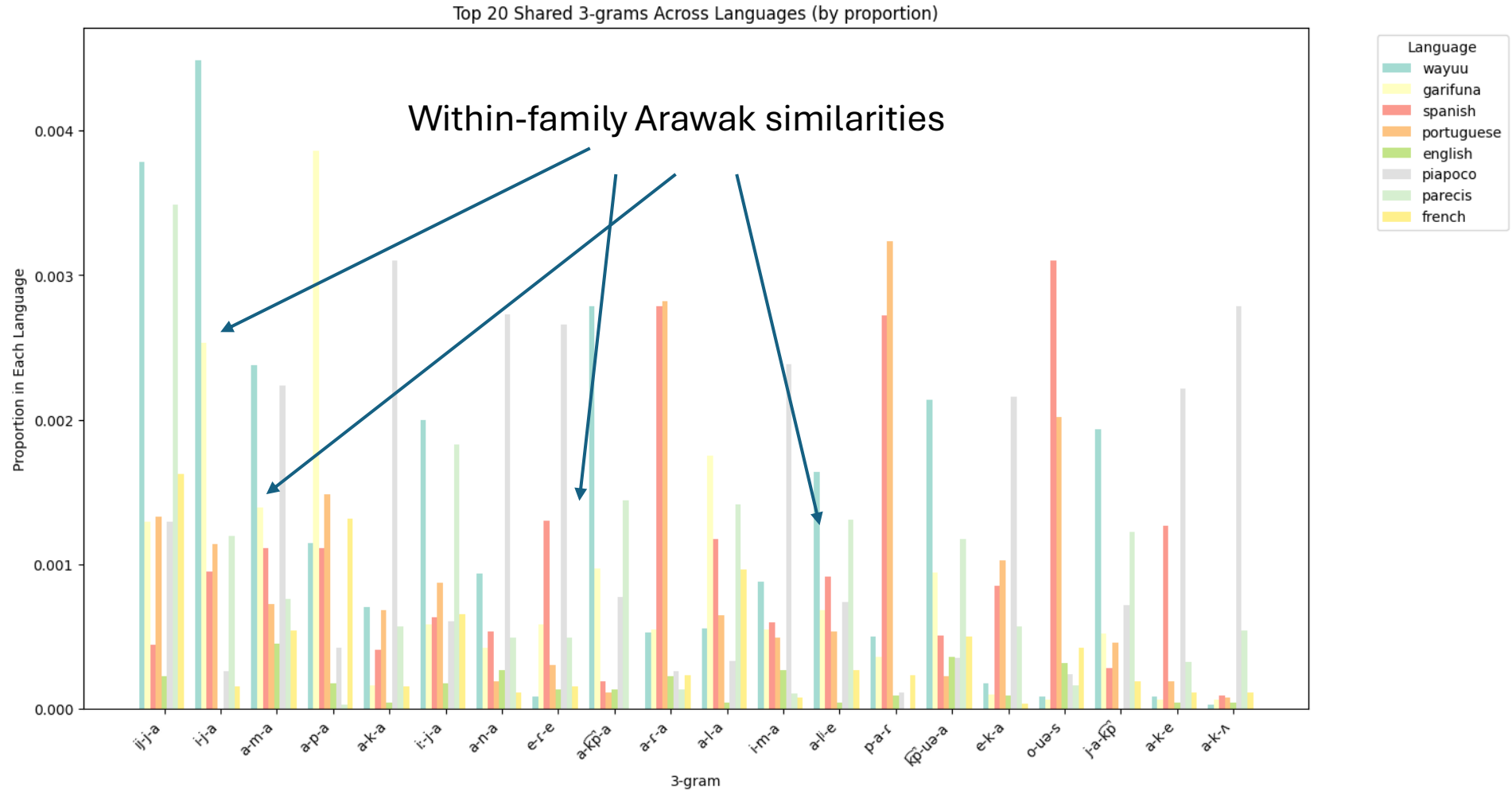
Bigram distributions



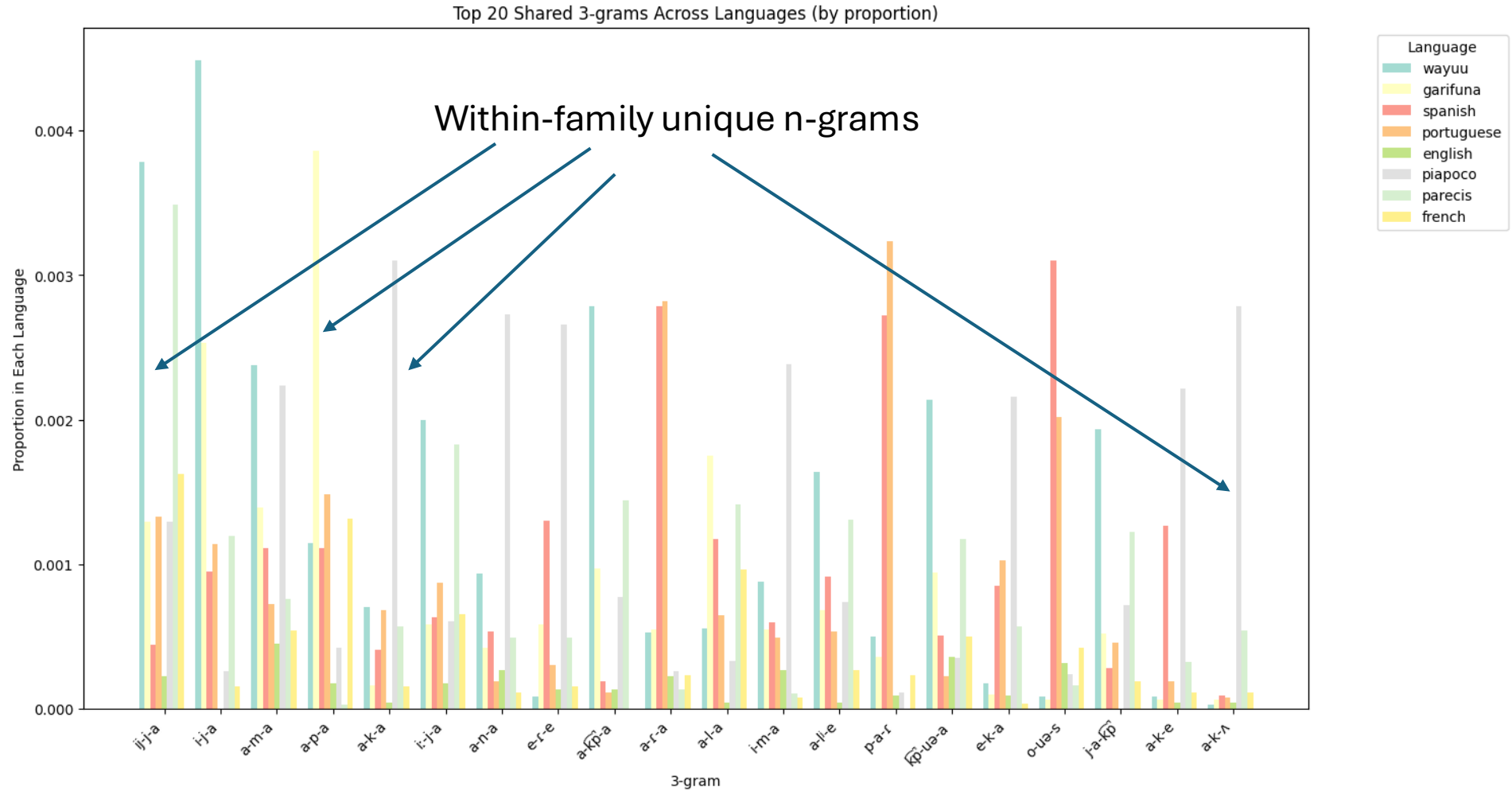
Bigram distributions



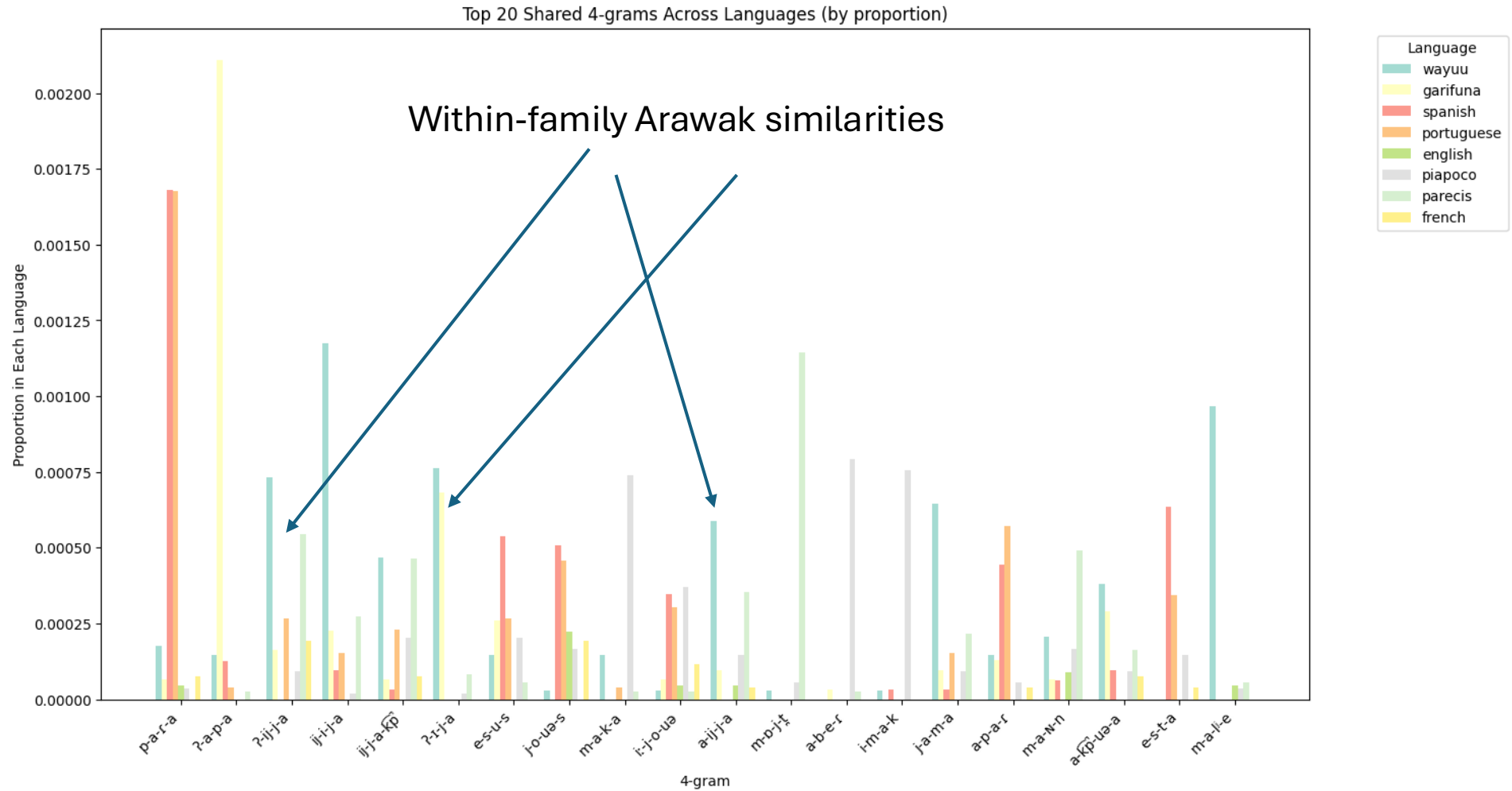
Trigram distributions



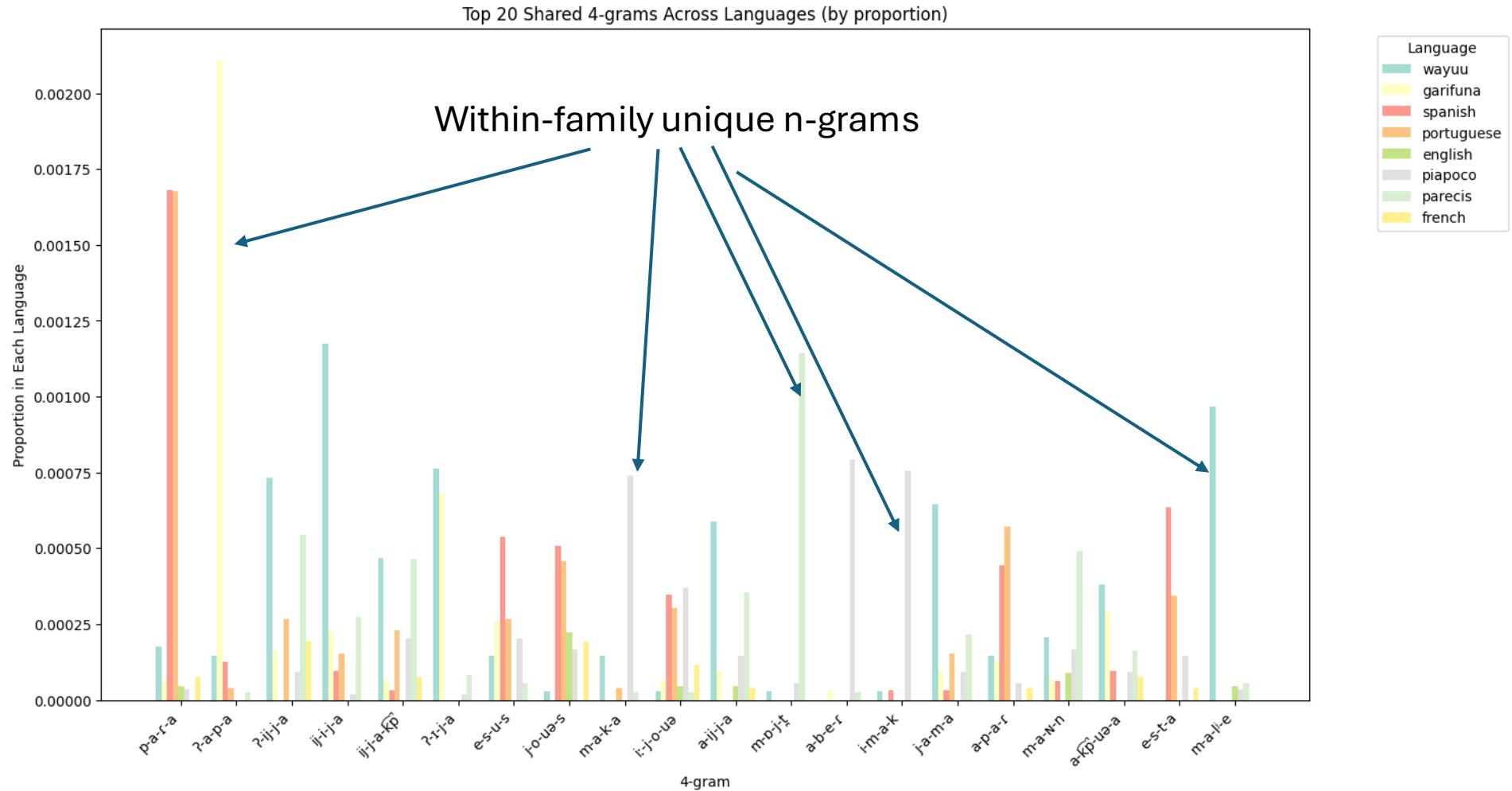
Trigram distributions



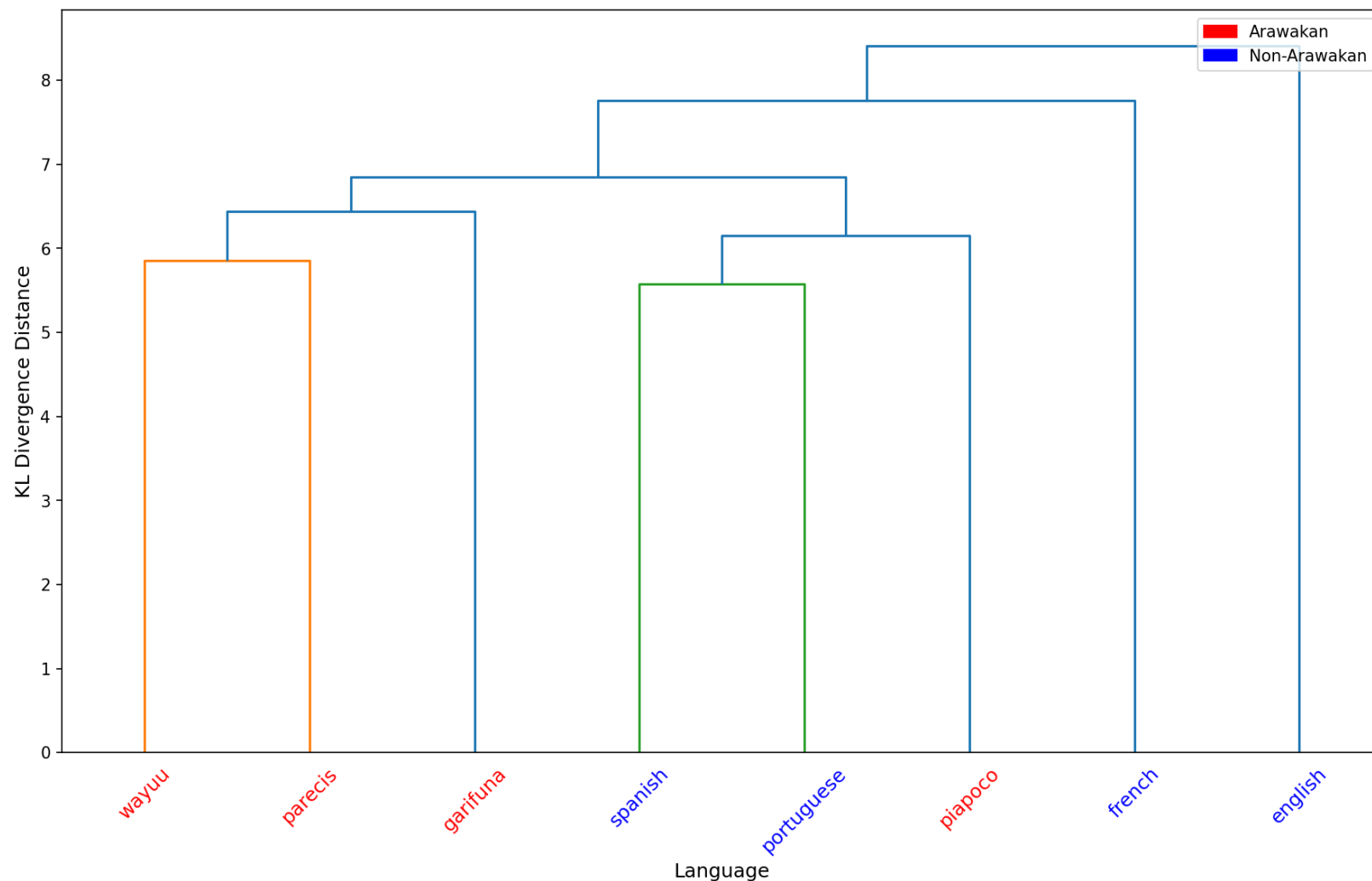
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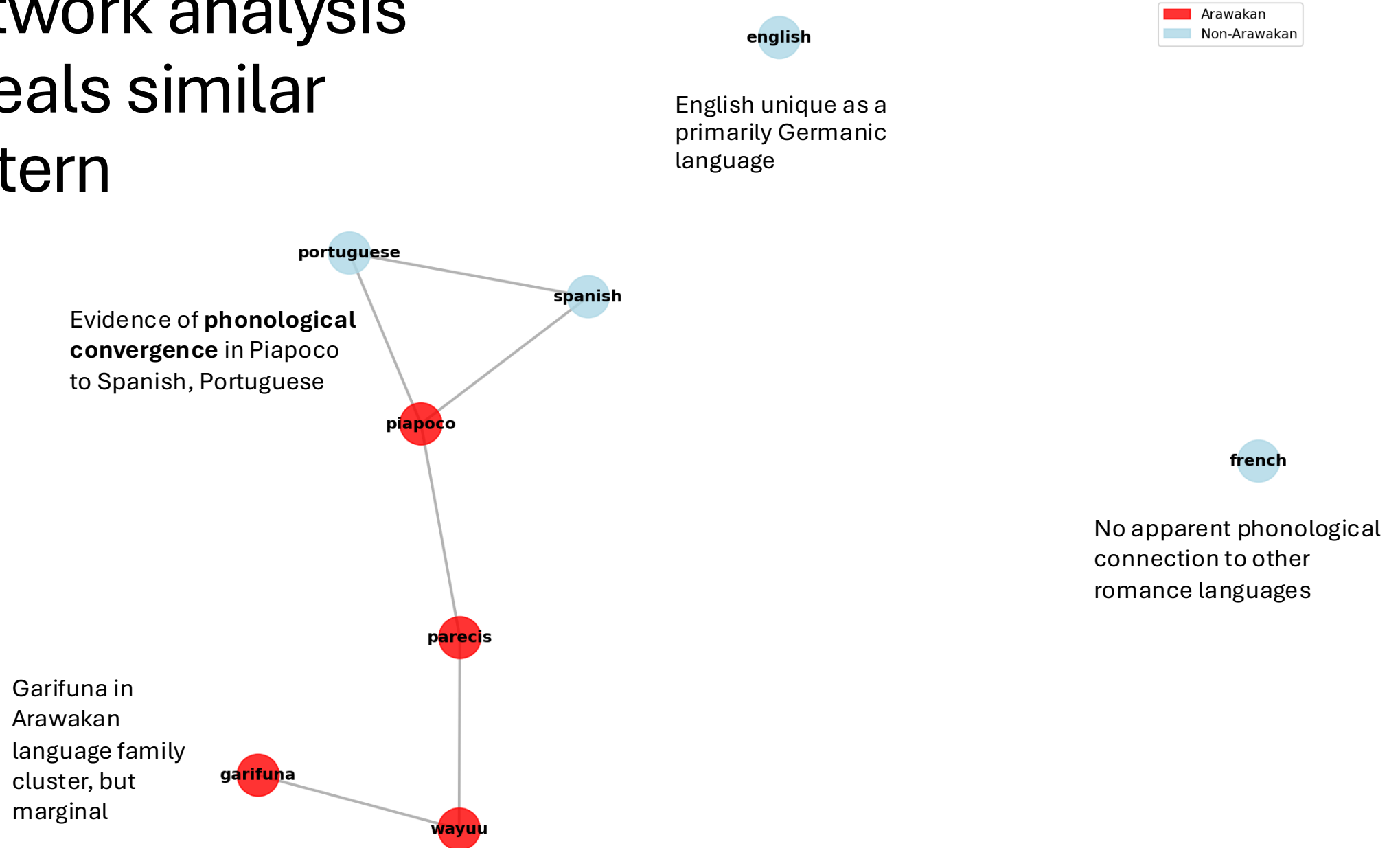
Quadgram distributions



Distributions cluster by mix of language family, historical external influences



Network analysis reveals similar pattern

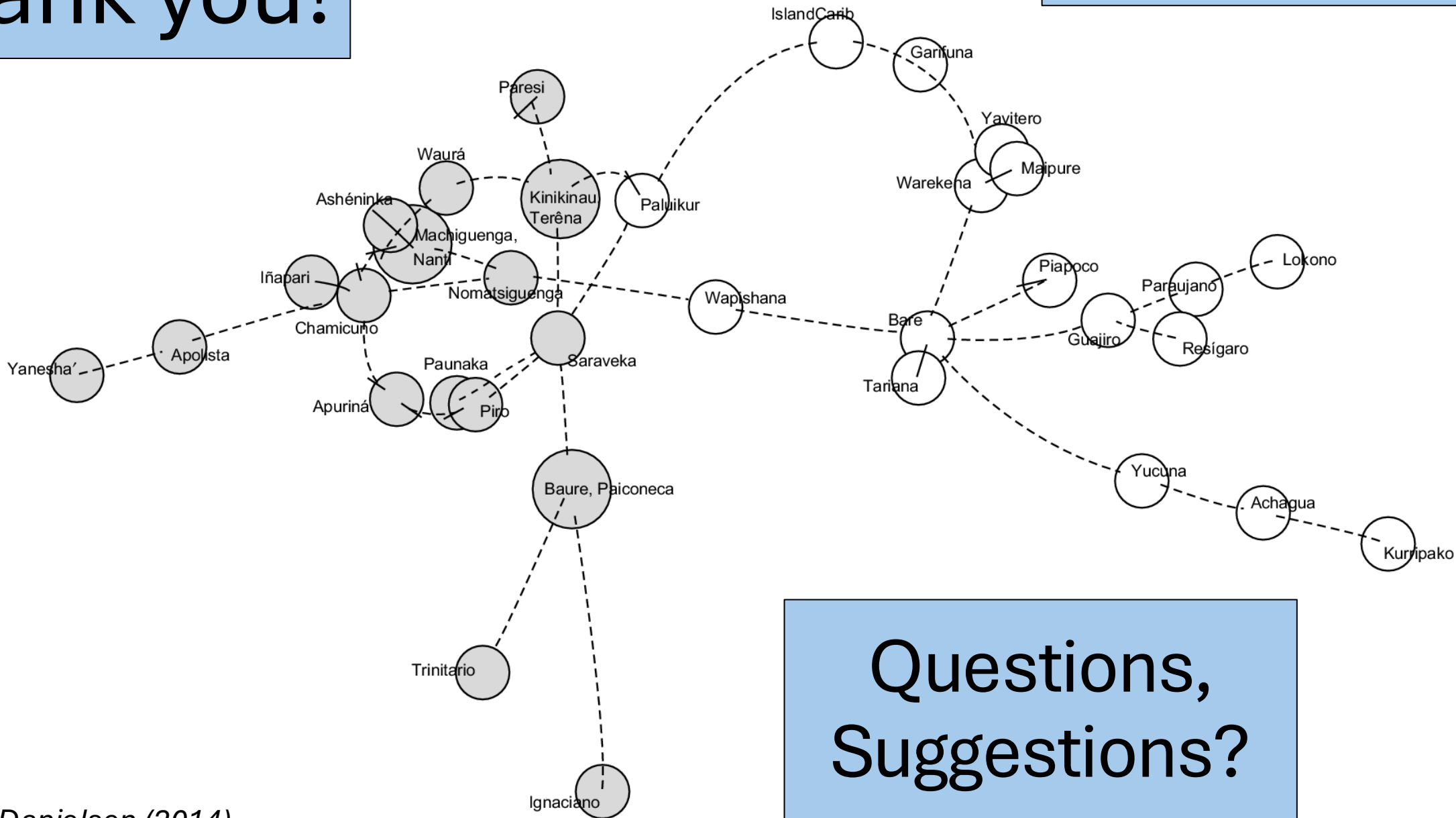


Conclusions

- Phone recognition tools reveal underlying shared phonological structure in Arawak language family
 - Ways to **connect** languages with similar roots
- Phonology captures possible geographical influences
- Future directions to study African influence in Garifuna phonology

Thank you!

Gratitude to ITRC and
Symposium Committee



Questions,
Suggestions?

Threshold selection for network representation

